

BUS713 Financial Analytics: Module 6

Feasibility Analysis: Break-Even, NPV, Forecasting, and Simulation

Modules 1 to 5 gave you tools to measure return, risk, required return, and hedging instruments. Module 6 provides you with the tools to help make decisions every business eventually faces: is a new project worth doing? You will build a small feasibility study end to end, then apply the same forecasting and simulation methods to the KO data you already know. Work through each task in order; later tasks reuse figures from earlier ones.

Task 1: Cost Structure, Break-Even, and Profit Projection

Scenario: A friend is launching a boutique meal-kit subscription business and has asked you, as the numbers person, whether it can realistically turn a profit.

Contribution margin is selling price minus variable cost per unit. Break-even quantity is fixed costs divided by contribution margin. Profit = Revenue - Variable costs - Fixed costs.

1. The business sells subscription boxes at \$75 each, with a variable cost of \$28 per box, and annual fixed costs of \$42,000. Calculate the contribution margin and the break-even quantity in R.
2. Write an R function `profit(q, p, v, f)` that returns profit for a given quantity, price, variable cost, and fixed cost. Use it to calculate profit at 800 boxes and at 1,000 boxes sold per year.
3. Your friend is confident they can sell 1,000 boxes but admits demand could realistically fall as low as 800. Based on your break-even quantity and the two profit figures, how much room for error does this business actually have? Would you call this business fragile or robust?

Task 2: NPV, IRR, and Real Options

Scenario: The meal-kit business has outgrown its founder's kitchen. She is now considering a \$250,000 investment in a small production facility and wants to know if it is worth it.

NPV discounts every future cash flow back to today's dollars and subtracts the initial investment; a positive NPV adds value. IRR is the discount rate that makes NPV equal zero. When NPV and IRR conflict, NPV should generally be preferred.

4. The facility requires an initial investment of \$250,000 and is expected to generate cash flows of \$70,000, \$85,000, \$95,000, \$60,000, and \$40,000 over the next five years. Using a discount rate of 9%, calculate the NPV in R with the FinCal package.
5. Calculate the IRR for the same cash flows. Is the project acceptable if the firm's required return is 9%? Would your conclusion change if the required return were 14%?
6. Suppose a conservative, static analysis of a related expansion project gives a negative NPV of -\$40,000. However, the project also carries an option to expand further if early sales are strong, worth an estimated \$90,000, and an option to abandon early if they are weak, protecting \$15,000 of downside value. Calculate the real-options-adjusted value of the project. Should it still be rejected on the basis of the static NPV alone?

Task 3: Sensitivity and Scenario Analysis

Scenario: Before recommending the meal-kit expansion, your friend's investor wants to see how profit responds if things do not go as planned.

Sensitivity analysis changes one variable at a time. Scenario analysis changes several variables together to reflect how conditions realistically move as a group.

7. Using the `profit()` function from Task 1, build a sensitivity table in R that calculates profit for quantities sold from 700 to 1,100 in steps of 100, holding price and variable cost at their original values. At what quantity does the business cross from loss into profit?
8. Build a three-row scenario table in R: a base case (price \$75, variable cost \$28, quantity 1,000), a demand-drop case (same price and cost, quantity 800), and a worst case (price \$70, variable cost \$32, quantity 800). Calculate profit for all three.
9. Compare the demand-drop case to the worst case. How much of the additional loss comes from lower volume alone, and how much comes from the added price and cost pressure? What would you tell the investor about which assumption the business is most exposed to?

Task 4: Forecasting KO's Share Price

Scenario: A colleague wants a purely mechanical, short-term forecast of KO's share price for an internal planning exercise. You know this is not investment advice, but it is a good test of the forecasting tools from this module.

A simple moving average is a useful benchmark for stable series but lags behind a trend. ETS chooses the best exponential smoothing variant automatically and can capture both trend and seasonality. A linear regression trend forecast is interpretable but assumes the trend is a straight line.

10. Download KO's monthly adjusted price and convert it to a `ts()` object. Fit a simple moving average with `ma(order = 12)`, forecast 12 months ahead with `forecast()`, and plot it.
11. Fit an `ets()` model to the same series, forecast 12 months ahead, and plot it. Compare the two forecasts: which one follows the recent trend more closely, and why, given how each method is built?
12. Fit a linear trend forecast with `tslm(price ~ trend)` and compare its `accuracy()` output to the ETS model's. Which has the lower RMSE? In 2 to 3 sentences, explain why extending a straight line forward to forecast a stock price is risky, even when the historical trend has been strongly upward. (Hint: think back to Module 2's discussion of why prices are non-stationary.)

Task 5: Monte Carlo Simulation of the Expansion Decision

Scenario: The investor from Task 3 is still not satisfied with a single NPV number for the production facility and wants to see the full range of possible outcomes before committing.

13. A smaller-scale version of the expansion requires an initial investment of \$180,000 over a 4-year horizon, with a discount rate of 8%. Annual revenue is expected to average \$120,000 with a standard deviation of \$15,000, and annual costs are expected to average \$60,000 with a standard deviation of \$8,000. Calculate the base case NPV in R using the expected values only.
14. Run a Monte Carlo simulation with 1,000 draws, using `rnorm()` for revenue and cost in each year, following the structure from this module. Report the mean simulated NPV, the 5th and 95th percentile outcomes, and the percentage of simulations that are viable ($NPV > 0$).
15. Plot a histogram of the simulated NPVs with a vertical line at zero. In 2 to 3 sentences, explain to the investor why this distribution is more useful for their decision than the single base case NPV number from Question 1.

Task 6: Writing the Feasibility Report

Scenario: You now have everything needed to write the feasibility section of the report: break-even, NPV and IRR, sensitivity results, and a Monte Carlo distribution. Your friend needs to present this to the investor next week.

16. Using your results from Tasks 1, 2, and 5, write a 5 to 7 sentence executive summary for the feasibility report, ending with a clear recommendation.
17. Identify the single biggest risk driver from your Task 3 sensitivity work or your Task 5 simulation. Explain, in plain language a non-technical investor could understand, why it matters more than the other assumptions.
18. The course material's example recommendation is conditional: the project should proceed "if marketing assumptions are achievable and costs remain within forecast." Write a similarly conditional recommendation sentence for the meal-kit expansion, based on what you actually found in Tasks 1 to 5.

Wrap-Up

You calculated break-even and profit under different assumptions, evaluated a project using NPV, IRR, and real options, built sensitivity and scenario tables, forecast a real price series with three different methods, quantified project uncertainty with Monte Carlo simulation, and wrote a feasibility recommendation grounded in all of it. This closes the loop from Module 1's first return calculation to a full investment decision.